SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1703**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

*I _**D.VAISHNAVI(192524143)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

ANSWER:

The knowledge base contains three rules and facts about Patient A.

Given

1. Fever AND Cough → Possible Flu
2. Fever AND Rash → Possible Measles
3. Cough AND Breathlessness → Possible Pneumonia
4. Patient A:
   - Fever = True
   - Cough = True
   - Breathlessness = True

(a) Represent using Propositional Logic

Define symbols

Let:

- F = Patient has Fever
- C = Patient has Cough
- B = Patient has Breathlessness
- R = Patient has Rash
- Flu = Patient has Possible Flu
- M = Patient has Possible Measles
- P = Patient has Possible Pneumonia

Rules

Rule 1:

$F \land C \rightarrow Flu$

Rule 2:

$F \land R \rightarrow M$

Rule 3:

$C \land B \rightarrow P$

Facts

F=True C=True B=True

There is no fact saying Rash is True.
Final representation
$(F \land C) \rightarrow Flu$ $(F \land R) \rightarrow M$ $(C \land B) \rightarrow P$ F, C, B

(b) Forward Chaining
Meaning
Forward Chaining starts with the known facts and moves forward by firing rules whose conditions are satisfied.
Step 1
Known facts:
F=True, C=True, B=True
Check Rule 1:
$F \land C \rightarrow Flu$
Both Fever and Cough are true.
Therefore:
Flu=True
Rule 1 fires → Possible Flu is derived.

Step 2
Check Rule 2:
$F \land R \rightarrow M$
We know:
F=True
But:
R=False/Unknown
Therefore Rule 2 cannot fire.
So:
M cannot be derived

Step 3
Check Rule 3:
$C \land B \rightarrow P$
We know:
C=True B=True
Therefore:
Pneumonia=True

Final answer:
Patient A has Possible Flu and Possible Pneumonia.
Measles cannot be concluded because Rash is not given.

(c) Backward Chaining – Prove Pneumonia
Goal
Pneumonia
Backward chaining starts with the goal and works backward to find the required conditions.
Step 1 – Find rule producing Pneumonia
Rule 3:
$C \land B \rightarrow P$
Therefore, to prove Pneumonia, we need:
Cough
and
Breathlessness

Step 2 – Check the facts

Knowledge base contains:

Cough=True

and

Breathlessness=True

Both conditions are satisfied.

Goal reduction tree

```
            Pneumonia
               ↑
        Cough AND Breathlessness
          /          \
         /            \
   Cough = True    Breathlessness = True
      ✓                ✓
```

Therefore:

Pneumonia=True

Conclusion

Patient A has Possible Pneumonia.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. ∀x: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x) [Rule 1]

II. ∀x: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x) [Rule 2]

III. ∀x ∀y: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

ANSWER:

Given Rules

Rule 1

∀xVehicle(x)∧EmergencyType(x)→ClearPath(x)

Rule 2

∀xClearPath(x)→GreenSignal(x)∧¬RedSignal(x)

Rule 3

∀x∀yVehicle(x)∧Behind(x,y)∧GreenSignal(y)→Proceed(x)

Facts

Vehicle(Ambulance) EmergencyType(Ambulance) Vehicle(CarA) Behind(CarA,Ambulance)

(a) Unification

We need to match Rule 1 with the facts.

Rule 1:

Vehicle(x)∧EmergencyType(x)→ClearPath(x)

We have:

Vehicle(Ambulance)

and

EmergencyType(Ambulance)

Therefore substitute:
$\theta = \{x/Ambulance\}$
After substitution:
$Vehicle(Ambulance) \land EmergencyType(Ambulance)$
Both facts are true.
Therefore:
$ClearPath(Ambulance)$
MGU

$\theta = \{x \mapsto Ambulance\}$

(b) Resolution – Prove Proceed(CarA)
We need to prove:
$Proceed(CarA)$
Step 1 – Convert Rule 1 to CNF
$Vehicle(x) \land EmergencyType(x) \rightarrow ClearPath(x)$
Using:
$A \rightarrow B \equiv \neg A \lor B$
we get:
$\neg Vehicle(x) \lor \neg EmergencyType(x) \lor ClearPath(x)$

Step 2 – Rule 2
Rule 2:
$ClearPath(x) \rightarrow GreenSignal(x) \land \neg RedSignal(x)$
This gives two clauses:
$\neg ClearPath(x) \lor GreenSignal(x)$ $\neg ClearPath(x) \lor \neg RedSignal(x)$

Step 3 – Rule 3
$Vehicle(x) \land Behind(x,y) \land GreenSignal(y) \rightarrow Proceed(x)$
CNF:
$\neg Vehicle(x) \lor \neg Behind(x,y) \lor \neg GreenSignal(y) \lor Proceed(x)$

Step 4 – Derive ClearPath(Ambulance)
Facts:
$Vehicle(Ambulance)$ $EmergencyType(Ambulance)$
Using Rule 1:
$ClearPath(Ambulance)$

Step 5 – Derive GreenSignal(Ambulance)
Using:
$\neg ClearPath(x) \lor GreenSignal(x)$
and:
$ClearPath(Ambulance)$
Resolution gives:
$GreenSignal(Ambulance)$

Step 6 – Derive Proceed(CarA)
Facts:
$Vehicle(CarA)$ $Behind(CarA, Ambulance)$
and derived fact:
$GreenSignal(Ambulance)$
Rule 3 therefore gives:
$Proceed(CarA)$
Resolution chain:

Vehicle(Ambulance)
+
EmergencyType(Ambulance)
↓
ClearPath(Ambulance)
↓
GreenSignal(Ambulance)
+
Vehicle(CarA)
+
Behind(CarA,Ambulance)
↓
Proceed(CarA)
Hence:
Proceed(CarA) is true.

(c) Two simultaneous emergency vehicles
The current KB can handle multiple vehicles individually, because the rules use variables.
For example:
Vehicle(Ambulance1) EmergencyType(Ambulance1)
gives:
ClearPath(Ambulance1)
Similarly:
Vehicle(Ambulance2) EmergencyType(Ambulance2)
gives:
ClearPath(Ambulance2)
However, additional facts/rules are needed for safe simultaneous routing.
For example:
Behind(CarA,Ambulance1) Behind(CarB,Ambulance2)
and emergency priority rules such as:
$EmergencyType(x) \land EmergencyType(y) \rightarrow PriorityControl(x,y)$
A conflict-resolution rule could also be added to determine which emergency vehicle gets priority if both require the same path.

Conclusion
The existing KB supports individual emergency vehicles, but for properly managing simultaneous emergency vehicles, it should include vehicle relationships, path conflicts, priority and traffic coordination rules.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: SoilDry → IrrigationNeeded
2) P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
3) P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

ANSWER:

The given propositional KB is:

1.SoilDry→IrrigationNeeded

2. IrrigationNeeded∧CropWheat→ApplyDripMethod

3.¬ApplyDripMethod∧CropWheat→CropAtRisk

Facts:
SoilDry=True CropWheat=True

(a) Convert to CNF

P1

Original:
SoilDry→IrrigationNeeded
Eliminate implication:
¬SoilDry∨IrrigationNeeded
This is already CNF.

P2

Original:
IrrigationNeeded∧CropWheat→ApplyDripMethod
Using:
A→B≡¬A∨B
where:
A=IrrigationNeeded∧CropWheat
we get:
¬(IrrigationNeeded∧CropWheat)∨ApplyDripMethod
Using De Morgan's law:
¬IrrigationNeeded∨¬CropWheat∨ApplyDripMethod
Therefore:
¬IrrigationNeeded∨¬CropWheat∨ApplyDripMethod

P3

Original:
¬ApplyDripMethod∧CropWheat→CropAtRisk
Eliminate implication:
¬(¬ApplyDripMethod∧CropWheat)∨CropAtRisk

Move negation inward:
ApplyDripMethod∨¬CropWheat∨CropAtRisk
Therefore:
ApplyDripMethod∨¬CropWheat∨CropAtRisk

Facts in CNF
SoilDry CropWheat
Final CNF
¬SoilDry∨IrrigationNeeded ¬IrrigationNeeded∨¬CropWheat∨ApplyDripMethod ApplyDripMethod∨¬CropWheat∨CropAtRisk SoilDry CropWheat

(b) Resolution Refutation – Prove ApplyDripMethod
Goal
ApplyDripMethod
For resolution refutation, negate the goal:
¬ApplyDripMethod
Add it to the KB.

Step 1
We have:
SoilDry
and:
¬SoilDry∨IrrigationNeeded
Resolve:
IrrigationNeeded

Step 2
We have:
IrrigationNeeded CropWheat
and:
¬IrrigationNeeded∨¬CropWheat∨ApplyDripMethod
Resolve with IrrigationNeeded:
¬CropWheat∨ApplyDripMethod
Resolve with CropWheat:
ApplyDripMethod

Step 3 – Contradiction
We initially added:
¬ApplyDripMethod
Now we have:
ApplyDripMethod
Therefore:
ApplyDripMethod∧¬ApplyDripMethod
gives the empty clause:
□
Resolution proof tree
```
          SoilDry
            +
   ¬SoilDry ∨ IrrigationNeeded
            ↓
      IrrigationNeeded
            +
          CropWheat
```

+
$\neg$IrrigationNeeded ∨ $\neg$CropWheat
∨ ApplyDripMethod
↓
ApplyDripMethod
+
$\neg$ApplyDripMethod
↓
□
Therefore:
ApplyDripMethod is proved.

(c) Remove SoilDry
Now remove:
SoilDry
from the KB.
Step 1
Rule P1 is:
SoilDry→IrrigationNeeded
But we no longer have the fact:
SoilDry
Therefore we cannot derive:
IrrigationNeeded

Step 2
P2 requires:
IrrigationNeeded∧CropWheat
Although:
CropWheat=True
we do not know:
IrrigationNeeded=True
Therefore:
ApplyDripMethod
cannot be derived.

Step 3 – CropAtRisk
P3 requires:
¬ApplyDripMethod∧CropWheat
We know:
CropWheat=True
But we do not have:
¬ApplyDripMethod
Not deriving ApplyDripMethod is not the same as proving ¬ApplyDripMethod.
Therefore:
CropAtRisk does not follow from the KB.
Final conclusion
After removing SoilDry:

- IrrigationNeeded → Cannot be derived
- ApplyDripMethod → Cannot be derived
- ¬ApplyDripMethod → Not given
- CropAtRisk → Cannot be derived

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1) Stench [1,2] → Wumpus is adjacent to cell [1,2]
2) Breeze [1,1] → Pit is adjacent to cell [1,1]
3) Glitter [x,y] → Gold is at [x,y]
4) ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

ANSWER:

The agent perceives:

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

The given rules are listed in the question paper.

(a) Construct Belief State and Apply Modus Ponens

Percepts

Stench[1,2] Breeze[1,1] Glitter[2,2]

Step 1 – Stench

Rule:

Stench[1,2]→WumpusAdjacent[1,2]

Given:

Stench[1,2]

Therefore, by Modus Ponens:

WumpusAdjacent[1,2]

Step 2 – Breeze

Rule:

Breeze[1,1]→PitAdjacent[1,1]

Given:

Breeze[1,1]

Therefore:

PitAdjacent[1,1]

Step 3 – Glitter

Rule:

Glitter[x,y]→Gold[x,y]

Given:

Glitter[2,2]

Therefore:

Gold[2,2]

Derived knowledge

Percept	Rule applied	Conclusion
Stench[1,2]	Stench → Wumpus adjacent	Wumpus is adjacent to [1,2]
Breeze[1,1]	Breeze → Pit adjacent	Pit is adjacent to [1,1]
Glitter[2,2]	Glitter → Gold	Gold is at [2,2]

Thus the belief state contains:
WumpusAdjacent[1,2] PitAdjacent[1,1] Gold[2,2]

(b) Forward Chaining – Possible Wumpus and Pit Locations

Wumpus

From:
Stench[1,2]
we infer:
WumpusAdjacent[1,2]
So the Wumpus can be in a cell adjacent to [1,2].
In a standard 4-directional grid, the adjacent cells are:

- [1,1]
- [1,3]
- [2,2]

So possible Wumpus locations are:
[1,1],[1,3],[2,2]
assuming these cells exist in the considered grid.

Pit

From:
Breeze[1,1]
we infer:
PitAdjacent[1,1]
The adjacent cell within the usual grid is:
[1,2]
Therefore the possible Pit location is:
[1,2]

Gold

From:
Glitter[2,2]
we directly obtain:
Gold[2,2]

(c) Is the path [1,1] → [1,2] → [2,2] safe?
The proposed path is:
[1,1] → [1,2] → [2,2]
We know:
Breeze[1,1]
Therefore:
PitAdjacent[1,1]
So [1,2] may contain a Pit.
We also know:
Stench[1,2]
Therefore:
WumpusAdjacent[1,2]

So a Wumpus may be located in one of the cells adjacent to [1,2].
Most importantly, the KB does not provide enough information to prove:
¬Pit[1,2]
or
¬Wumpus[1,2]
Therefore, the cell [1,2] cannot be guaranteed safe.

Final answer

The path cannot be confirmed as safe.
The agent should not assume the path is safe, because the available knowledge indicates possible danger around the path.
Also, although:
Gold[2,2]
is confirmed, reaching it safely is not logically guaranteed from the given KB.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand